Efficient Open Domination and Exact Total Domination in Cylindrical Grids

박성현

Abstract

We characterize efficient open domination in every cylindrical grid $P_m \square C_n$ with $m \geq 2$ and $n \geq 3$. Such a set exists if and only if $4 \mid n$ for odd m, and $m+1 \mid n$ for even m. The proof uses a tridiagonal row-sum system for necessity and an explicit symbolic residue construction for even-width sufficiency. The same system gives a general lower bound for total domination. In particular, for all $k, t \geq 1$, $\gamma_t(P_{2k} \square C_{(2k+1)t}) = k(k+1)t$. We then complement these structural results with exact total-domination classifications for widths five, six, and seven. Those residual fixed-width results are certified by independent finite-state and min-plus matrix identities, including the width-six transient exception.

Keywords: total domination; cylindrical grid; Cartesian product; finite-state method; min-plus algebra; computer-assisted proof.

1 Introduction

Let G be a graph without isolated vertices. A set $D \subseteq V(G)$ is *totally dominating* if every vertex of G, including every vertex of D, has a neighbor in D. The minimum cardinality of such a set is the total domination number $\gamma_t(G)$. In this paper we study the cylindrical grids $P_m \square C_n$, the Cartesian product of a path and a cycle.

Our first result concerns the stronger exact condition in which every vertex has exactly one selected open neighbor. We call such a set an efficient open dominating set (EOD set); the same object is also called a total perfect code or an exact transversal. The main structural theorem is

Theorem 1.1 (All-width EOD characterization)**.** For $m \geq 2$ and $n \geq 3$, $P_m \square C_n$ has an efficient open dominating set if and only if

$$\big(m \text{ odd and } 4 \mid n\big) \quad \text{or} \quad \big(m \text{ even and } m+1 \mid n\big).$$

The theorem is proved by summing the exact open-neighbor equations over each cycle row. If s_i is the number of selected vertices in row i, then

$$2s_i + s_{i-1} + s_{i+1} = n, \qquad s_0 = s_{m+1} = 0.$$

For odd width this forces all even rows to be empty and imposes a period-four horizontal pattern. For even width it forces divisibility by $m+1$; the converse is supplied by a symbolic construction on $P_{2k} \square C_{2k+1}$, lifted periodically. The proof includes $m = 2, n = 3$ and the path-boundary rows.

The same tridiagonal system yields, for every total dominating set,

$$\gamma_t(P_m \square C_n) \geq \begin{cases} \lceil (m+1)n/4 \rceil, & m \text{ odd}, \\ \lceil m(m+2)n/(4(m+1)) \rceil, & m \text{ even}. \end{cases}$$

The even-width EOD construction attains the bound whenever $n = (2k+1)t$, giving the exact two-parameter family

$$\gamma_t(P_{2k} \,\square\, C_{(2k+1)t}) = k(k+1)t.$$

Total domination in grid-like products has a substantial literature. Gravier studied total domination in products of paths and cycles and related tiling formulations [6]. Hu, Sohn, and Chen record exact values for $P_p \,\square\, C_3$ and $P_p \,\square\, C_4$ for every $p \geq 2$ [8]. Eakawinrujee gives exact values for path widths $p = 2, 3, 4$ and $q \geq 5$ [3]; the author's dissertation gives additional provenance for that cylinder program [4]. Rectangular grids, ordinary domination, and several variants provide nearby but distinct problems; for example, the fixed-width results of Crevals and Ostergard concern $P_m \,\square\, P_n$, not cylinders [2], while Nandi, Parui, and Adhikari study ordinary domination [13].

The standard reference by Haynes, Hedetniemi, and Henning includes efficient total domination and a dedicated discussion of total efficient cylindrical graphs [7]. We use that treatment as background and distinguish its recorded cylindrical results from the all-width characterization proved here.

Related toroidal-mesh calculations of Hu and Xu are also distinct from the path-boundary cylinders studied here [9]. Recent work on proper total domination likewise concerns a different domination variant [14].

The results below concern cases not covered by the fixed-width cylindrical formulas cited above. Hu, Sohn, and Chen also study related bundle and toroidal cases, including C_m bundles over C_n and an exact result for $C_n \square C_5$; those results are not silently identified with the cylinder $P_5 \,\square\, C_n$ [8]. Eakawinrujee's public dissertation records the exact subfamily

$$p \equiv 1 \pmod 2, \quad q \equiv 0 \pmod 4 \quad \Longrightarrow \quad \gamma_t(P_p \,\square\, C_q) = \frac{(p+1)q}{4},$$

which overlaps the present width-five and width-seven formulas on infinitely many circumferences [4]. We therefore make the narrow claim of complete classifications for widths five, six, and seven across all $n \geq 3$ in the audited record, extending rather than erasing these earlier exact subfamilies. The cyclic boundary condition couples the first and last columns, and a selected vertex cannot dominate itself. Our approach handles both issues with a finite-state representation; the semantic reduction is proved in the paper and the finite identities are supplied as exact computer-assisted certificates.

The methodological distinction from recent nearby work is also important. Garzón, Martínez, Moreno, and Puertas already use min-plus algorithms for 2-domination of small-cycle cylinders [5]. Martínez, Castaño-Fernández, and Puertas later use tropical matrix products for broader 2-domination cylinder bounds [12]; that invariant requires two selected neighbors for vertices outside the set, whereas ordinary total domination requires one open neighbor for every vertex, including selected vertices. The shared min-plus language is methodological context, not an equivalence of parameters.

1.1 Prior results and scope of the contribution

The relevant boundary of the literature can be summarized as follows. The first five rows concern the same total-domination or EOD parameters, with the fixed variable shown explicitly; the last two provide nearby context. The EOD entries refer specifically to Proposition 5.6 and Theorem 5.7 of Kuziak–Peterin–González Yero [11].

category/source	parameter	scope relevant to this paper
Known EOD / Kuziak–Peterin–González Yero (2014)	EOD	small circumferences $C_3, \dots, C_7$ (Thm. 5.7), and the odd family $P_{2r+1} \square C_{4t}$ (Prop. 5.6)
Known exact / Hu–Sohn–Chen (2016)	total domination	$P_p \square C_3$ and $P_p \square C_4$, all $p \geq 2$, plus bundle/toroidal cases
Known exact / Eakawinrujee (2022)	total domination	$P_p \square C_q$ with $p = 2, 3, 4$, $q \geq 5$, and related bounds
Known provenance / dissertation	total domination	$q = 3, 4$ split and odd p, $q \equiv 0 \pmod 4$ subfamily
New structural theorem / this paper	EOD	iff: odd m, $4 \mid n$, or even m, $m+1 \mid n$, all $m \geq 2, n \geq 3$
New exact family / this paper	total domination	$\gamma_t(P_{2k} \square C_{(2k+1)t}) = k(k+1)t$, all $k, t \geq 1$
New residual classifications / this paper	total domination	exact values for every $n \geq 3$ at widths $5, 6, 7$, including finite transients
Nearby method / Garzón–Martínez–Moreno–Puertas (2021)	2-domination	min-plus algorithms for small-cycle cylinders; a different invariant
Nearby method / Martínez–Castaño-Fernández–Puertas (2024)	2-domination	finite-state/tropical methods for cylinders; a different invariant

The table intentionally separates prior fixed circumferences from prior fixed path widths, and separates both from the all-width claims made here.

Our main results are the following.

Theorem 1.2 (Width five)**.** For every $n \geq 3$,

$$\gamma_t(P_5 \square C_n) = \begin{cases} \lceil 3n/2 \rceil + 1, & n \equiv 2 \pmod 4, \\ \lceil 3n/2 \rceil, & \text{otherwise.} \end{cases}$$

Theorem 1.3 (Width six)**.** Let ε_r, for $r \in \{0, \dots, 13\}$, be given by

$$\varepsilon_r = \begin{cases} 0, & r \in \{0, 3, 5, 7, 9, 10\}, \\ 1, & r \in \{1, 4, 6, 11, 13\}, \\ 2, & r \in \{2, 8\}, \\ 3, & r = 12. \end{cases}$$

Then

$$\gamma_t(P_6 \square C_n) = \left\lceil \frac{12n}{7} \right\rceil + \varepsilon_{n \bmod 14}$$

for every $n \geq 3$, except that $\gamma_t(P_6 \square C_{12}) = 22$.

Theorem 1.4 (Width seven)**.** For every $n \geq 3$,

$$\gamma_t(P_7 \square C_n) = 2n,$$

except $\gamma_t(P_7 \square C_7) = 15$ and $\gamma_t(P_7 \square C_{14}) = 30$.

The proof has two layers. First, we prove the graph-theoretic equivalence between total dominating sets and weighted closed walks. Second, a finite certificate establishes the matrix identities and prefix values. The thresholded recurrence lemma then shows exactly how those finite facts imply each all-circumference formula.

For orientation, exact small-circumference cylinder cases are already part of the prior record discussed by Hu–Sohn–Chen and reproduced in the public Eakawinrujee dissertation. For the three target widths, those prior C_3- and C_4-values are $5, 6, 6, 8, 8, 6, 8$, respectively. The same values are independently present in our finite prefix, but are not claimed as new. The table separates these checked small circumferences from the all-circumference classification supplied below; it does not attribute the cylinder cases to Hu–Sohn–Chen's separate toroidal $C_n \square C_5$ result:

m	$\gamma_t(P_m \,\square\, C_3)$	$\gamma_t(P_m \,\square\, C_4)$
5	5	6
6	6	8
7	6	8

The common row operator. The same tridiagonal operator provides the conceptual bridge between the two parts of the paper: EOD gives the equality $B_m s = n\mathbf{1}$, whereas an arbitrary total dominating set gives the componentwise inequality $B_m d \geq n\mathbf{1}$. We use this equality–inequality pair to obtain the general lower bound before turning to the residual fixed-width cases.

2 Efficient open domination and the structural lower bound

We give the proof of Theorem 1.1 and the lower bound used later. Write $x_{i,j} = 1$ when (i, j) belongs to an EOD set and 0 otherwise, with j read modulo n. The defining condition is

$$x_{i-1,j} + x_{i+1,j} + x_{i,j-1} + x_{i,j+1} = 1, \tag{1}$$

where path indices outside $\{1, \ldots, m\}$ contribute zero. Put $s_i = \sum_j x_{i,j}$, $s_0 = s_{m+1} = 0$. Summing (1) over j gives

$$2s_i + s_{i-1} + s_{i+1} = n. \tag{2}$$

Here B_m denotes the $m \times m$ tridiagonal matrix with diagonal entries 2 and entries 1 immediately above and below the diagonal; thus (2) is $B_m s = n\mathbf{1}$. The solution is unique: for $z_0 = z_{m+1} = 0$,

$$z^\mathsf{T} B_m z = \sum_{i=0}^{m} (z_i + z_{i+1})^2 > 0$$

for every nonzero z, so B_m is positive definite.

Lemma 2.1 (The row solution)**.** If $m = 2k + 1$, the unique solution of (2) is $s_i = n/2$ for odd i and $s_i = 0$ for even i. If $m = 2k$, then $s_i = na_i/(2k + 1)$, where

$$a_{2r-1} = k - r + 1, \qquad a_{2r} = r$$

whenever the indices lie in $\{1, \ldots, k\}$, and $a_{2k+1-i} = a_i$.

Proof. Substitution verifies both displayed solutions, including the two boundary equations, and uniqueness was proved above. In the even case the value 1 occurs among the a_i, so integrality of s_i forces $2k + 1 \mid n$. □

For odd m, every even row is empty. On an odd row, (1) becomes $x_{i,j-1} + x_{i,j+1} = 1$, hence $x_{i,j+2} = 1 - x_{i,j}$ and $x_{i,j+4} = x_{i,j}$. If n is odd, the step $j \mapsto j + 2$ is one cycle, so the relation would force $x_{i,j} = 1 - x_{i,j}$, impossible for a Boolean value (equivalently, the row count $s_i = n/2$ is not integral). Thus n is even; after $n/2$ two-steps the value must agree with its starting value, so $n/2$ is even and $4 \mid n$. Conversely, when $4 \mid n$, put alternating period-four patterns $1100\ldots$ and $0011\ldots$ on successive odd rows and put no vertices on even rows. Every position in either

1100 or 0011 has exactly one selected horizontal neighbor; each interior even-row vertex has exactly one of its two vertical neighbors selected, and the path-boundary rows are odd. The pattern closes because $4 \mid n$. This is an EOD set.

For completeness, the even converse is recorded as a symbolic residue lemma. It is the only construction-specific calculation in the all-width theorem. Let $N = 2k+1$, $E_0 = \varnothing$, and

$$\widehat{E}_r = \{\, 2(r-1) - 4t : 0 \le t < r \,\}, \qquad E_r = \{\, e \bmod N : e \in \widehat{E}_r \,\}.$$

If $k = 2h$, set $A_{2r} = E_r$ for $1 \le r \le h$, set

$$A_{2r+1} = E_r \cup C_r, \qquad \text{where } C_r = \bigcup_{t=0}^{h-r-1} \{2r+2+4t, 2r+3+4t\} \quad (0 \le r < h), \qquad C_h = \varnothing,$$

and reflect by $A_{2k+1-i} = A_i$. If $k = 2h+1$, set $A_{2r} = E_r$ for $1 \le r \le h$, set

$$A_{2r-1} = E_r \cup D_r, \qquad \text{where } D_r = \bigcup_{t=0}^{h-r} \{2r+1+4t, 2r+2+4t\} \quad (1 \le r \le h+1),$$

and apply the same reflection. Empty unions are understood literally. In the interval calculation below, $\widehat{E}_r$ denotes the displayed integer representatives; all resulting set identities are reduced modulo N, while C_r and D_r use their displayed integer representatives.

Lemma 2.2 (Even-width residue construction)**.** With nonexistent boundary sets interpreted as empty, these sets satisfy

$$A_{i-1} \,\dot\cup\, A_{i+1} \,\dot\cup\, (A_i - 1) \,\dot\cup\, (A_i + 1) = \mathbb{Z}_N \tag{3}$$

for every $1 \le i \le 2k$. Moreover $|A_i| = a_i$.

Proof. We give the residue calculation explicitly. For an integer interval write $[a, b]_{\mathbb{Z}} = \{a, a+1, \ldots, b\}$, and put

$$U_r = E_{r-1} \cup E_r \cup (E_r - 1) \cup (E_r + 1), \qquad V_r = E_r \cup E_{r+1} \cup (E_r - 1) \cup (E_r + 1).$$

Grouping the integer representatives of $\widehat{E}_r$ with the same t gives four consecutive integers $2r-4-4t, \ldots, 2r-1-4t$, and the final three shifted residues are $1-2r, 2-2r, 3-2r$. Hence, before reduction modulo N,

$$U_r = [1-2r, 2r-1]_{\mathbb{Z}}. \tag{4}$$

The corresponding calculation for V_r has one additional endpoint:

$$V_r = [1-2r, 2r]_{\mathbb{Z}} \cup \{-2r\}. \tag{5}$$

These identities are used only in ranges where the displayed endpoints have length at most N, so the subsequent wrap-around is explicit.

Suppose first $k = 2h$, so $N = 4h+1$. The boundary row identity is

$$E_1 \dot\cup (C_0 - 1) \dot\cup (C_0 + 1) = \mathbb{Z}_N, \tag{6}$$

because $E_1 = \{0\}$ and the two translates of C_0 are the blocks $[1,4]_{\mathbb{Z}}, [5,8]_{\mathbb{Z}}, \ldots, [4h-3, 4h]_{\mathbb{Z}}$. For $1 \le r \le h$, C_{r-1} and C_r concatenate to $[2r, 4h-2r+1]_{\mathbb{Z}}$; together with (4) this proves

$$(E_{r-1} \cup C_{r-1}) \dot\cup (E_r \cup C_r) \dot\cup (E_r - 1) \dot\cup (E_r + 1) = \mathbb{Z}_N. \tag{7}$$

For $1 \le r < h$, $C_r - 1$ and $C_r + 1$ concatenate to $[2r+1, 4h-2r]_{\mathbb{Z}}$. The interval part of (5) then gives $[1-2r, 4h-2r]_{\mathbb{Z}}$, and its next residue modulo N is $-2r$, proving

$$E_r \dot\cup E_{r+1} \dot\cup ((E_r \cup C_r) - 1) \dot\cup ((E_r \cup C_r) + 1) = \mathbb{Z}_N. \tag{8}$$

The $r = 0$ instance is (6), and the central row is (7) at $r = h$.

Now suppose $k = 2h + 1$, so $N = 4h + 3$. For $1 \le r \le h$, $D_r - 1$ and $D_r + 1$ concatenate to $[2r, 4h - 2r + 3]_{\mathbb{Z}}$; with (4) this proves

$$E_{r-1} \dot\cup E_r \dot\cup ((E_r \cup D_r) - 1) \dot\cup ((E_r \cup D_r) + 1) = \mathbb{Z}_N. \tag{9}$$

For $r = h + 1$, $D_r = \varnothing$ and (4) itself has N elements, so (9) also covers the upper boundary. For $1 \le r \le h$, D_r and D_{r+1} concatenate to $[2r + 1, 4h - 2r + 2]_{\mathbb{Z}}$; the interval part of (5) plus the single residue $-2r$ proves

$$(E_r \cup D_r) \dot\cup (E_{r+1} \cup D_{r+1}) \dot\cup (E_r - 1) \dot\cup (E_r + 1) = \mathbb{Z}_N. \tag{10}$$

Reflection $i \mapsto 2k + 1 - i$ exchanges the two neighboring row sets, so (6)–(10) cover every reflected row as well. Counting the disjoint pieces gives

$$|A_{2r}| = r, \quad |A_{2r+1}| = 2h - r \quad (k = 2h),$$

and

$$|A_{2r}| = r, \quad |A_{2r-1}| = 2h - r + 2 \quad (k = 2h + 1);$$

reflection supplies the other rows. The central and boundary cases are the omitted empty-union cases; for $k = 1, N = 3$, (9) is the direct three-column identity. Thus every row satisfies (3). $\square$

Taking $D = \{(i, j) : j \in A_i\}$, identity (3) says exactly that every vertex has one open neighbor in D. Repeating the pattern with period N proves the even sufficiency whenever $N \mid n$, and the odd construction above proves the other sufficiency. Together with the row solution and its divisibility consequences, this proves Theorem 1.1.

We next derive the total-domination bound. If D is any total dominating set and $d_i = |D \cap (\{i\} \times C_n)|$, summing the domination inequalities over row i gives

$$2d_i + d_{i-1} + d_{i+1} \ge n. \tag{4}$$

The vector $y = B_m^{-1}\mathbf{1}$ is nonnegative (the explicit solutions in Lemma 2.1 give it directly); it is strictly positive when m is even. Since $y^{\mathsf{T}} B_m = \mathbf{1}^{\mathsf{T}}$, multiplying (4) by y_i and summing gives the explicit dual chain

$$|D| = \mathbf{1}^{\mathsf{T}} d = y^{\mathsf{T}} B_m d \ge n\, y^{\mathsf{T}} \mathbf{1} = \begin{cases} (m+1)n/4, & m \text{ odd}, \\ m(m+2)n/[4(m+1)], & m \text{ even}. \end{cases} \tag{5}$$

Since $|D|$ is integral, taking ceilings gives the stated lower bound. For $m = 2k$, the construction in Lemma 2.2 has $\sum_i |A_i| = \sum_i a_i = k(k+1)$. Its period-$2k+1$ lift to $C_{(2k+1)t}$ preserves (3), including the seam, and has size $k(k+1)t$. Equality in (5) therefore proves

$$\gamma_t(P_{2k} \,\square\, C_{(2k+1)t}) = k(k+1)t.$$

3 Preliminaries

We use $[m] = \{0, 1, \ldots, m-1\}$ for the rows of P_m, and index the cycle columns by $\mathbb{Z}/n\mathbb{Z}$. For $S \subseteq [m]$, define its open path neighborhood by

$$N_P(S) = \{i \in [m] : i - 1 \in S \text{ or } i + 1 \in S\},$$

where indices outside $[m]$ are omitted. Thus a row selected in a column does not automatically dominate the vertex in that same row and column.

The preceding EOD section used one-based path-row labels $1, \ldots, m$. For the finite-state encoding below we relabel the same path vertices by $[m] = \{0, \ldots, m-1\}$; this is only a shift of row labels and does not change any adjacency or domination condition.

For a set $D \subseteq V(P_m \,\Box\, C_n)$, write

$$S_j = \{i \in [m] : (i,j) \in D\}.$$

The horizontal neighbors of (i,j) are $(i,j-1)$ and $(i,j+1)$, and the vertical neighbors are the vertices in rows adjacent to i in the same column.

4 The finite-state model

For a cyclic sequence of selected row sets (S_j), define the pending set

$$R_j = [m] \setminus \big(S_{j-1} \cup N_P(S_j)\big).$$

The rows in R_j are precisely those for which the left horizontal and vertical domination alternatives are absent in column j. Define the fixed admissible state set

$$Q_m = \{(S,R) : S \subseteq [m],\ R \subseteq [m] \setminus N_P(S),\ \text{and } R = [m] \setminus (U \cup N_P(S)) \text{ for some } U \subseteq [m]\}.$$

Thus the predecessor mask is quantified in the definition but is not part of the state. Given a state $q = (S,R) \in Q_m$ and an input mask $T \subseteq [m]$, the transition is legal precisely when

$$R \subseteq T,$$

and its head pending set is forced to be

$$R' = [m] \setminus \big(S \cup N_P(T)\big).$$

The edge is $(S,R) \to (T,R')$, where the displayed formula defines R', provided $R \subseteq T$ and $(T,R') \in Q_m$; its weight is $|T|$. Every edge head is therefore a member of the fixed finite set Q_m.

Theorem 4.1 (Finite-state representation)**.** For $m \in \{5,6,7\}$ and $n \geq 3$, labeled total dominating sets of $P_m \,\Box\, C_n$ are in weight-preserving bijection with labeled length-n closed walks in the state graph above. The weight of a walk is the sum of its transition weights.

Proof. We first prove soundness. Let a cyclic legal walk be given and define $D = \{(i,j) : i \in S_j\}$. Consider a vertex (i,j). If $i \notin R_j$, then by the definition of R_j, either $i \in S_{j-1}$ or $i \in N_P(S_j)$. Hence (i,j) has a selected left or vertical open neighbor. If $i \in R_j$, transition legality gives $i \in S_{j+1}$, so the right horizontal neighbor is selected. These cases include selected vertices themselves and therefore establish total domination.

Conversely, let D be a total dominating set and define S_j from its selected vertices and R_j by the displayed formula. If $i \in R_j$, then the left and every vertical selected-neighbor alternative is absent. Total domination forces the right neighbor $(i,j+1)$ to be selected, so $R_j \subseteq S_{j+1}$. The next pending formula follows from its definition. The same argument applies to the wraparound transition.

The two constructions are inverse because the pending masks are forced by the selected masks. Finally,

$$\sum_j |S_{j+1}| = \sum_j |S_j| = |D|,$$

so weights agree. For $n = 3$, the left and right horizontal neighbors are distinct because $3 \nmid 2$, so the cyclic encoding has exactly the simple graph neighborhood semantics. $\square$

4.1 A local transition example

It is useful to see the pending-mask convention in a concrete transition. Take $S = \varnothing$ and $R = [m]$. This is an admissible state: before the current column is read, every row may still require its right horizontal neighbor. Choose $T = [m]$. The edge is legal because $R \subseteq T$, and its head pending set is

$$[m] \setminus \big(\varnothing \cup N_P([m])\big) = \varnothing.$$

The transition has weight m. In the corresponding two-column fragment, the selected vertices in T are dominated vertically, while the rows that were pending in the tail receive their required right neighbor. This example also illustrates why the state records obligations created by the preceding column rather than treating a selected vertex as self-dominating.

More generally, the state condition is not an additional graph restriction. For $R \subseteq [m] \setminus N_P(S)$, taking

$$U = [m] \setminus \big(R \cup N_P(S)\big)$$

gives $R = [m] \setminus (U \cup N_P(S))$. Thus the quantified predecessor mask in the definition of Q_m is exactly the realizability condition for the residual obligation mask. The independent state reconstruction uses this equivalent characterization.

5 Min-plus path semantics and the recurrence certificate

Let M_m be the weighted adjacency matrix of this graph over $\mathbb{Z}_{\geq 0} \cup \{+\infty\}$, with $+\infty$ for a missing edge. Min-plus addition and multiplication are

$$a \oplus b = \min(a, b), \qquad a \otimes b = a + b,$$

with $+\infty$ absorbing for $\otimes$. Matrix powers use min-plus multiplication. The min-plus identity matrix has zero diagonal and $+\infty$ off the diagonal. For $c \in \mathbb{Z}_{\geq 0}$, scalar matrix shift means $(c \otimes A)_{uv} = c + A_{uv}$ for finite A_{uv}, and $(c \otimes A)_{uv} = +\infty$ otherwise.

Lemma 5.1 (Min-plus path semantics)**.** For every $k \geq 0$, $(M_m^k)_{uv}$ is the minimum weight of a length-k walk from u to v, with the minimum of an empty set interpreted as $+\infty$.

Proof. The case $k = 0$ is the min-plus identity matrix. For the induction step, partition a length-$k+1$ walk by its penultimate state x. The prefix has minimum cost $(M_m^k)_{ux}$, the final edge has cost $(M_m)_{xv}$, and minimizing over the finite state set gives the min-plus product. Missing prefixes or edges contribute $+\infty$ and therefore do not create a walk. $\square$

Define

$$g_m(n) = \min_{q \in Q_m} (M_m^n)_{qq}.$$

Theorem 4.1 and Lemma 5.1 give

$$\gamma_t(P_m \,\square\, C_n) = g_m(n), \qquad n \geq 3.$$

The distinction between an entrywise identity and a scalar sequence pattern is important. An identity for the minimum diagonal alone would not control arbitrary endpoints and would not justify concatenating an additional walk. The certificates instead compare every matrix entry at the two exponents. This is the reason the recurrence lemma applies uniformly to all closed walks, including those whose minimizing state changes with the circumference.

Lemma 5.2 (Thresholded min-plus recurrence)**.** Suppose an entrywise identity

$$M^{N+p} = c \otimes M^N$$

holds for a finite integer c. Define $h(t) = \min_q (M^t)_{qq}$. Then for every $k \geq 0$,

$$M^{N+p+k} = c \otimes M^{N+k},$$

and hence $h(n+p) = h(n) + c$ for every finite-value base $n \geq N$.

Proof. Min-plus matrix multiplication is associative: both parenthesizations minimize the same finite collection of triple products. A finite scalar shift commutes with a subsequent min-plus product, including entries equal to $+\infty$. Therefore, for $k \geq 0$,

$$M^{N+p+k} = M^{N+p} \otimes M^k = (c \otimes M^N) \otimes M^k = c \otimes (M^N \otimes M^k) = c \otimes M^{N+k}.$$

Taking the minimum over diagonal entries gives the same statement for g. When the relevant diagonal minimum is finite, the scalar operation is ordinary addition. The conclusion is explicitly restricted to $n = N + k \geq N$; no value below the threshold is propagated by this lemma. □

6 Finite certificates and exact formulas

The finite certificate proposition below is the computational input to the three theorem proofs. Its verifier reconstructs the state and transition semantics, parses the persisted matrices, recomputes both powers, and checks the entrywise identities with explicit $+\infty$ semantics. The proposition is computer-assisted: its trusted inputs are the definitions, the certificate files, and the verifier specified in the reproducibility record.

Proposition 6.1 (Finite certificate and prefix data)**.** For widths five, six, and seven, the state and transition graphs have the counts in Table 1; the displayed matrix identities hold; and the diagonal minima $g_m(n)$ agree with the exact finite-prefix tables in the project artifact on $3 \leq n \leq 19$, $3 \leq n \leq 34$, and $3 \leq n \leq 31$, respectively.

Table 1: Finite state graphs and entrywise recurrence certificates.

width	states	transitions	N	p	c
5	169	2,419	16	4	6
6	441	11,025	21	14	24
7	1,156	50,303	28	4	8

The certified identities are

$$M_5^{20} = 6 \otimes M_5^{16}, \qquad M_6^{35} = 24 \otimes M_6^{21}, \qquad M_7^{32} = 8 \otimes M_7^{28}.$$

Consequently the recurrences begin at $n = 16$, $n = 21$, and $n = 28$, respectively. The exact finite-prefix replay covers $3 \leq n \leq 19$ for width five, $3 \leq n \leq 34$ for width six, and $3 \leq n \leq 31$ for width seven. Since each interval ends at $N + p - 1$, every later index reduces to a checked base in $[N, N + p - 1]$.

The decisive finite values are displayed here. The complete machine-readable tables, including all rows from 3 through the stated prefix endpoint, are part of the reproducibility record.

Table 2: Width-five prefix values.

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$g_5(n)$	5	6	8	10	11	12	14	16	17	18	20	22	23	24	26	28	29

Table 3: Width-six base and exceptional values.

n	12	21	22	23	24	25	26	27	28	29	30	31	32	33	34
$g_6(n)$	22	36	40	40	42	44	48	48	48	51	54	54	56	57	60

Table 4: Width-seven exceptional and tail-base values.

n	7	14	28	29	30	31
$g_7(n)$	15	30	56	58	60	62

Complete finite-prefix tables

For auditability, the omitted prefix entries are listed here as well. These tables are the exact diagonal minima from the clean-room trace, not values inferred from the eventual recurrence.

Table 5: Complete width-six prefix, split into readable blocks.

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$g_6(n)$	6	8	9	12	12	16	16	18	20	22	24	24	27	30	30	32

n	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
$g_6(n)$	33	36	36	40	40	42	44	48	48	48	51	54	54	56	57	60

6.1 Width five

The identity $M_5^{20} = 6 \otimes M_5^{16}$ gives $g_5(n+4) = g_5(n) + 6$ for $n \geq 16$. *Proof of Theorem 1.2.* The finite proposition gives every value $3 \leq n \leq 19$; the base values $16 \leq n \leq 19$ are shown in Table 2, and the complete prefix comparison is in the machine-readable table. Every $n \geq 20$ reduces by four to one of these bases, and the residue $n \equiv 2 \pmod 4$ contributes the additional unit.

6.2 Width six

The identity $M_6^{35} = 24 \otimes M_6^{21}$ gives $g_6(n+14) = g_6(n)+24$ only for $n \geq 21$. *Proof of Theorem 1.3.* The value at $n = 12$ is a finite-prefix exception and is not propagated to $n = 26$. The finite proposition gives every value $3 \leq n \leq 34$, including all rows $3 \leq n < 21$ and every base value $21 \leq n \leq 34$. For every $n \geq 35$, repeated subtraction of fourteen reaches one of the checked values $21, \ldots, 34$, which proves Theorem 1.3; the recurrence is never applied below its threshold.

6.3 Width seven

The identity $M_7^{32} = 8 \otimes M_7^{28}$ gives $g_7(n+4) = g_7(n) + 8$ only for $n \geq 28$. *Proof of Theorem 1.4.* The finite proposition explicitly covers every row $3 \leq n \leq 31$, including the ordinary prefix rows and the exceptions $g_7(7) = 15$ and $g_7(14) = 30$. Every $n \geq 32$ reduces to a base in $28, \ldots, 31$, proving the theorem. The exceptions remain only in the finite prefix and are not propagated by the recurrence.

7 Verification and reproducibility

The repository includes a reference implementation and a separately written certificate verifier. The verifier rebuilds states, transitions, weights, and matrix powers from the definitions.

Table 6: Complete width-seven prefix, split into readable blocks.

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$g_7(n)$	6	8	10	12	15	16	18	20	22	24	26	30	30	32	34

n	18	19	20	21	22	23	24	25	26	27	28	29	30	31
$g_7(n)$	36	38	40	42	44	46	48	50	52	54	56	58	60	62

Detailed test counts, mutation results, certificate manifests, and hash bindings are recorded separately from the mathematical proof. The three certificate commands are:

```
python verify/verify_certificates.py 5 certificates/width5
python verify/verify_certificates.py 6 certificates/width6
python verify/verify_certificates.py 7 certificates/width7
```

Each command reports zero identity mismatches. The repository's SHA256SUMS file binds the finite artifacts, and paper/REPRODUCIBILITY.md records the expected dimensions, counts, parameters, and verification outputs.

The computations establish the finite proposition; the preceding semantic lemmas and theorem arguments establish why that proposition implies the infinite statements.

7.1 What is and is not certified

The finite proposition has three logically separate inputs. First, the definition-level state and transition reconstruction fixes the finite graph. Second, the persisted matrices are parsed with explicit finite and $+\infty$ tags, and the verifier recomputes both powers from the reconstructed edges rather than trusting a producer's claimed identity. Third, the prefix trace initializes every state as a possible source and minimizes the matching diagonal entry, so its values are the unrestricted $g_m(n)$, not a low-weight-source surrogate. Hashes bind the persisted binary artifacts to the manifests used by the verifier.

This separation also fixes the boundary of the computer-assisted argument. The code supplies finite exact premises; it does not replace the total-domination bijection, min-plus path lemma, or thresholded algebraic implication proved above. Conversely, the written argument does not assert that a finite set of tested values alone proves an infinite formula: the entrywise identities and their thresholds are the step that supplies the tail.

7.2 Data and code availability

The source code, finite certificates, independent verification software, and reproducibility materials are publicly available at `https://github.com/coshaman/cylinder`. No DOI is currently assigned to the code archive.

8 Discussion

The structural result comes first: exact open-neighborhood partitioning on a cylinder is governed by the tridiagonal operator B_m. The equality $B_m s = n\mathbf{1}$ forces row rigidity and parity-dependent divisibility, whereas the inequality $B_m d \geq n\mathbf{1}$ for an arbitrary total dominating set gives the universal lower bound. The explicit even-width construction attains that bound on $P_{2k} \square C_{(2k+1)t}$ for every $k, t \geq 1$.

The odd-width $4 \mid n$ EOD family is known; it is recorded by Kuziak–Peterin–González Yero [11]. The present structural contribution is the complete iff characterization, including the even-width branch, together with the sharp arbitrary-even-width total-domination family.

The recent torus result of Wehrmann and Koster [15] concerns $C_m \square C_n$, whereas this paper concerns $P_m \square C_n$; the cyclic and path boundaries make these distinct problems.

The width-five, width-six, and width-seven classifications provide residual information not determined by the EOD subsequences alone. Their eventual periods are four, fourteen, and four, respectively. The width-six EOD-sharp subsequence is $7 \mid n$, since $m + 1 = 7$, but the full period-fourteen correction is an independent certified property and is not claimed to follow from the EOD theorem alone. Values below the recurrence thresholds remain finite-prefix data.

Finite-state and min-plus transfer methods are established tools for repetitive graphs [10, 1]. Here they provide an exact, auditable completion layer for widths 5–7, not the primary methodological novelty. The state space grows rapidly with the width, so no arbitrary-width formula for $\gamma_t(P_m \square C_n)$, and no claim about width eight, is made. Natural future directions include sharper residual bounds and other boundary conditions.

9 Conclusion

We completely characterized efficient open domination in every cylindrical grid $P_m \square C_n$. The same tridiagonal row system then yielded a universal lower bound for total domination. The explicit construction makes that bound sharp for the arbitrary-even-width family $P_{2k} \square C_{(2k+1)t}$. We next completed the total-domination formulas for widths five, six, and seven. Only this residual fixed-width layer is computer-assisted: its total-domination/automaton bijection, min-plus path semantics, matrix identities, and threshold induction are explicit, and the finite premises are independently replayable.

References

[1] Yassine Bouznif, Julien Moncel, and Myriam Preissmann. A constant time algorithm for some optimization problems in rotagraphs and fasciagraphs. *Discrete Applied Mathematics*, 208:27–40, 2016.

[2] Simon Crevals and Patric R. J. Ostergard. Total domination of grid graphs. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 101:175–192, 2017.

[3] Pannawat Eakawinrujee. Total and paired domination numbers of cylinders. *Bulletin of the Malaysian Mathematical Sciences Society*, 45:3321–3334, 2022.

[4] Pannawat Eakawinrujee. *Total and paired domination numbers and gamma-total and gamma-paired dominating graphs of some graphs.* PhD thesis, Thammasat University, 2023.

[5] E. M. Garzon, J. A. Martinez, J. J. Moreno, and M. L. Puertas. On the 2-domination number of cylinders with small cycles. *Fundamenta Informaticae*, 185:185–199, 2022.

[6] Sylvain Gravier. Total domination number of grid graphs. *Discrete Applied Mathematics*, 121:119–128, 2002.

[7] Teresa W. Haynes, Stephen T. Hedetniemi, and Michael A. Henning. *Domination in Graphs: Core Concepts.* Springer Monographs in Mathematics. Springer, 2023.

[8] Fu-Tao Hu, Moo Young Sohn, and Xue-gang Chen. Total and paired domination numbers of C_m bundles over a cycle C_n. *Journal of Combinatorial Optimization*, 32:608–625, 2016.

[9] Fu-Tao Hu and Jun-Ming Xu. Total and paired domination numbers of toroidal meshes. *Journal of Combinatorial Optimization*, 27:76–88, 2014.

[10] Sandi Klavžar and Janez Žerovnik. Algebraic approach to fasciagraphs and rotagraphs. *Discrete Applied Mathematics*, 68(1–2):93–100, 1996.

[11] Dorota Kuziak, Iztok Peterin, and Ismael González Yero. Efficient open domination in graph products. *Discrete Mathematics and Theoretical Computer Science*, 16(1):105–120, 2014.

[12] Jose Antonio Martinez, Ana Belen Castano-Fernandez, and Maria Luz Puertas. The 2-domination number of cylindrical graphs. *arXiv preprint arXiv:2409.16703*, 2024.

[13] Mrinal Nandi, Subrata Parui, and Avishek Adhikari. The domination numbers of cylindrical grid graphs. *Applied Mathematics and Computation*, 217:4879–4889, 2011.

[14] Sawyer Osborn and Ping Zhang. From princes on chessboards to proper total domination in graphs. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 129:33–48, 2026.

[15] Moritz Wehrmann and Arie M. C. A. Koster. Efficient total domination and related invariants in torus graphs. *Discrete Applied Mathematics*, 389:106–118, 2026.